

Autonomous Research Agent

research-agent-python — LangChain + Groq + Streamlit

1 PROBLEM

- Deep research is slow and manual — hours spent searching, opening tabs, and cross-checking sources
- Generic chatbot answers lack ground-truth detail and traceability to real sources
- Search tools return raw links, not a structured, decision-ready report

2 SOLUTION

- Autonomous LangChain agent plans its own multi-step research from a single goal
- Calls web-search and page-scraping tools to gather ground-truth detail
- Groq-powered reasoning shown live; compiles a structured Markdown report
- Streamlit UI with adjustable iteration limit and one-click export

3 UNIQUE VALUE PROPOSITION

- Turn one plain-English research goal into a fully structured, cited report — with the agent's reasoning visible at every step
- The transparent, autonomous research analyst that shows its work
- No prompt engineering or manual searching — just describe the goal

8 UNFAIR ADVANTAGE

- Groq inference speed keeps multi-step agent loops fast enough to feel interactive
- Live, transparent reasoning builds trust versus black-box tools
- Open-source codebase invites community trust and contribution

6 CUSTOMER SEGMENTS

- Students & academic researchers running literature or topic scans
- Content marketers and writers needing sourced background research
- Analysts & consultants scoping a new topic or market
- Indie developers and founders validating ideas quickly
- Early adopters: AI/LangChain hobbyists and open-source users

9 EXISTING ALTERNATIVES

- Manual web research (search engines + note-taking)
- General-purpose chatbots without visible reasoning steps
- Paid research or analyst services

4 KEY METRICS

- Reports generated per week
- Weekly active users on the app
- Avg. tool calls per completed report
- Report download & repeat-use rate

5 CHANNELS

- Public GitHub repo and README
- Live demo on Streamlit Community Cloud
- Developer communities and LinkedIn
- Product Hunt, Hacker News, AI directories

7 COST STRUCTURE

- LLM inference costs via Groq API (per-token usage)
- Web search / scraping API or service fees

7 REVENUE STREAMS

- Freemium — free tier with limited runs, paid tier for higher iteration limits
- Usage-based API pricing for report generation